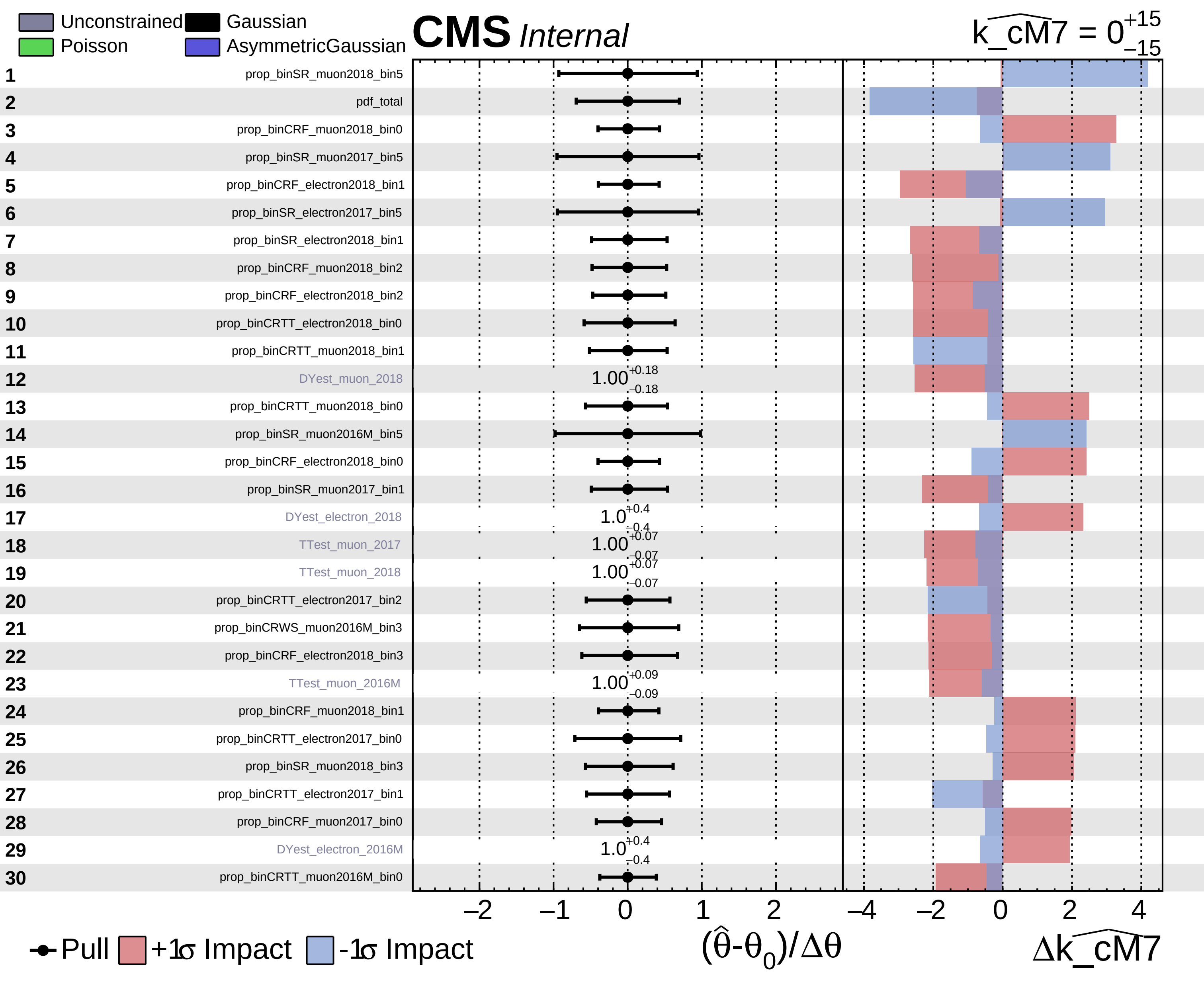
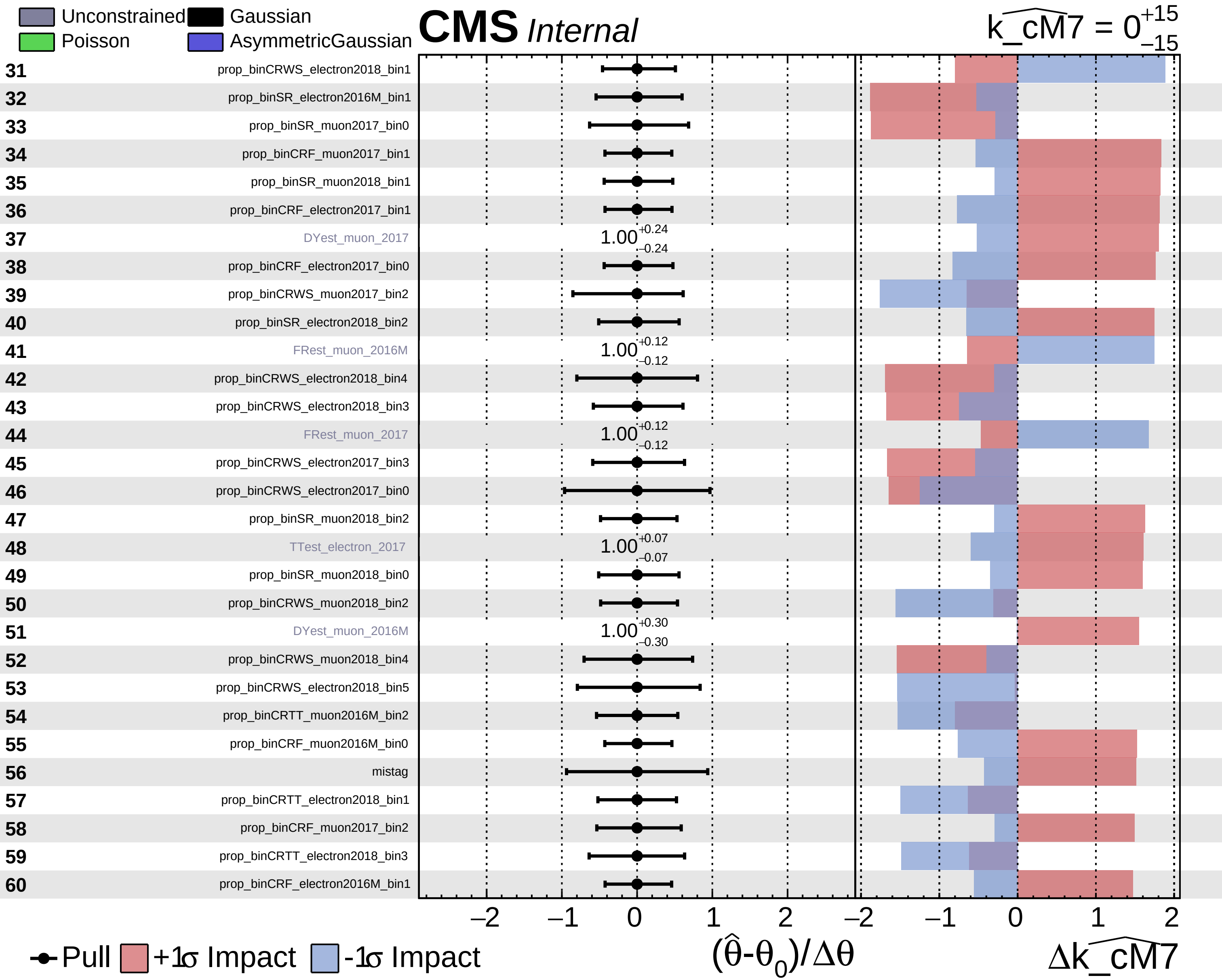
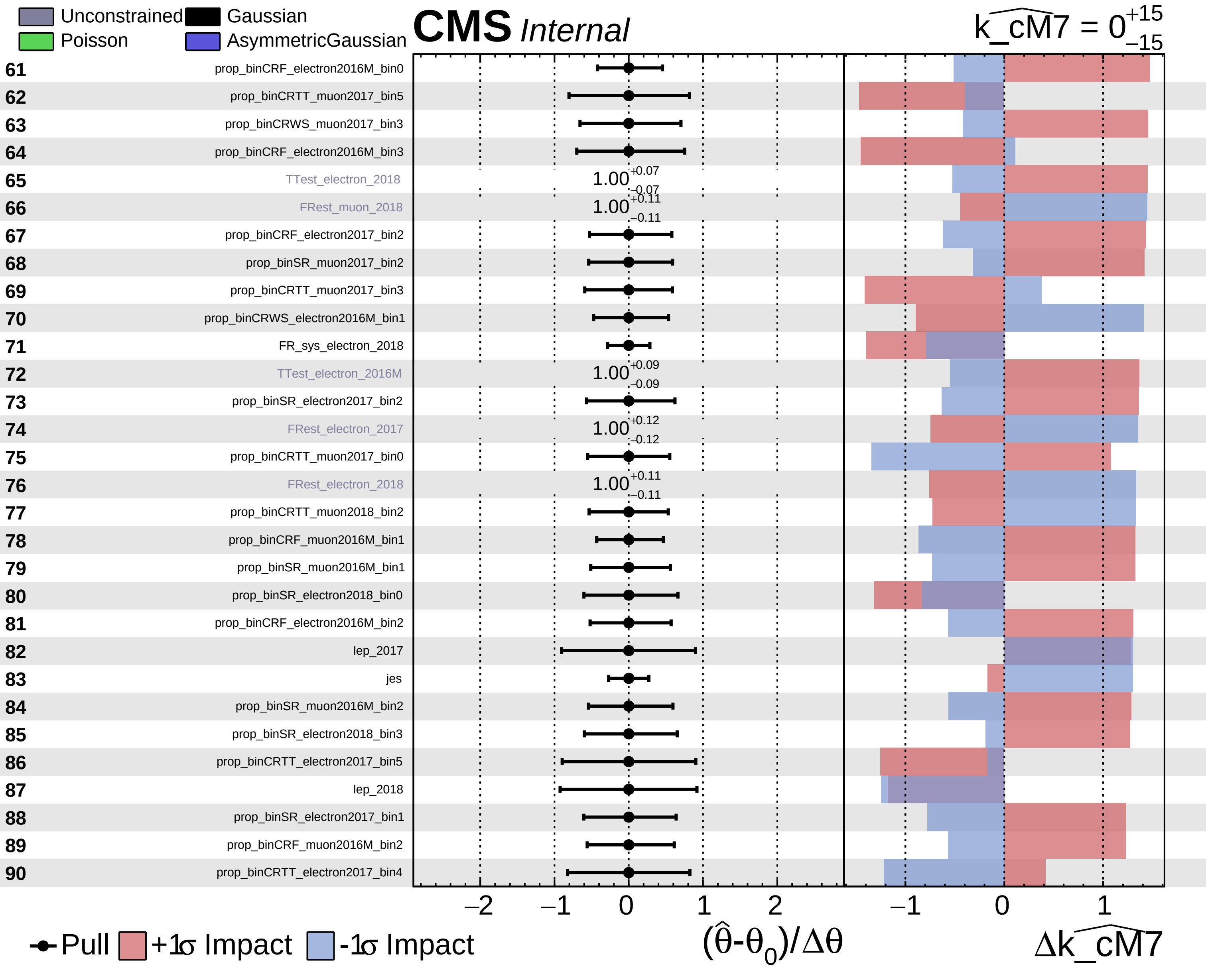


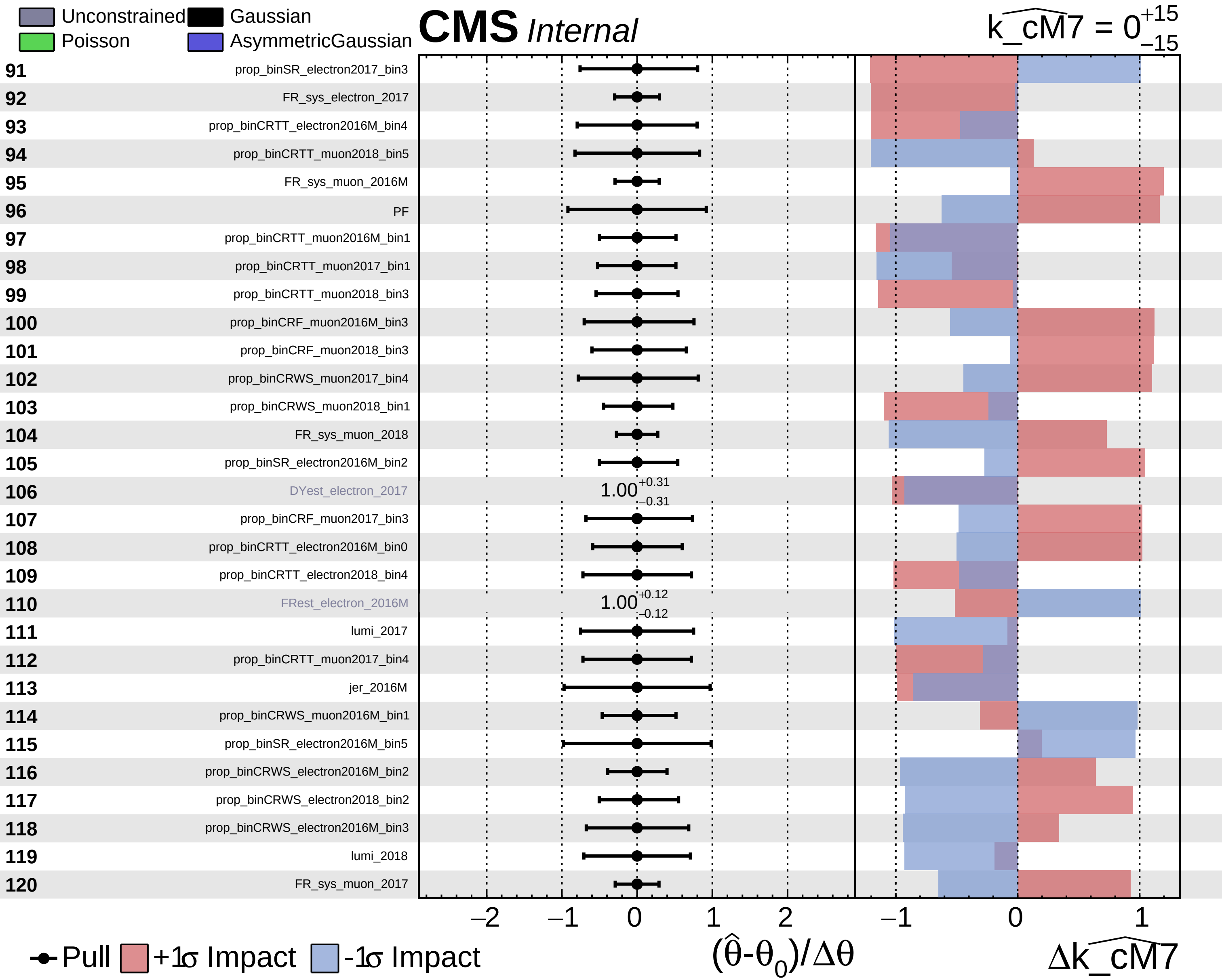
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+15}_{-15}$
1 prop_binSR_muon2018_bin5
2 pdf_total
3 prop_binCRF_muon2018_bin0
4 prop_binSR_muon2017_bin5
5 prop_binCRF_electron2018_bin1
6 prop_binSR_electron2017_bin5
7 prop_binSR_electron2018_bin1
8 prop_binCRF_muon2018_bin2
9 prop_binCRF_electron2018_bin2
10 prop_binCRTT_electron2018_bin0
11 prop_binCRTT_muon2018_bin1
12 DYest_muon_2018
$1.00^{+0.18}_{-0.18}$
13 prop_binCRTT_muon2018_bin0
14 prop_binSR_muon2016M_bin5
15 prop_binCRF_electron2018_bin0
16 prop_binSR_muon2017_bin1
17 DYest_electron_2018
$1.0^{+0.4}_{-0.4}$
18 TTest_muon_2017
$1.00^{+0.07}_{-0.07}$
19 TTest_muon_2018
$1.00^{+0.07}_{-0.07}$
20 prop_binCRTT_electron2017_bin2
21 prop_binCRWS_muon2016M_bin3
22 prop_binCRF_electron2018_bin3
23 TTest_muon_2016M
$1.00^{+0.09}_{-0.09}$
24 prop_binCRF_muon2018_bin1
25 prop_binCRTT_electron2017_bin0
26 prop_binSR_muon2018_bin3
27 prop_binCRTT_electron2017_bin1
28 prop_binCRF_muon2017_bin0
29 DYest_electron_2016M
$1.0^{+0.4}_{-0.4}$
30 prop_binCRTT_muon2016M_bin0
−2 −1 0 1 2
−4 −2 0 2 4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



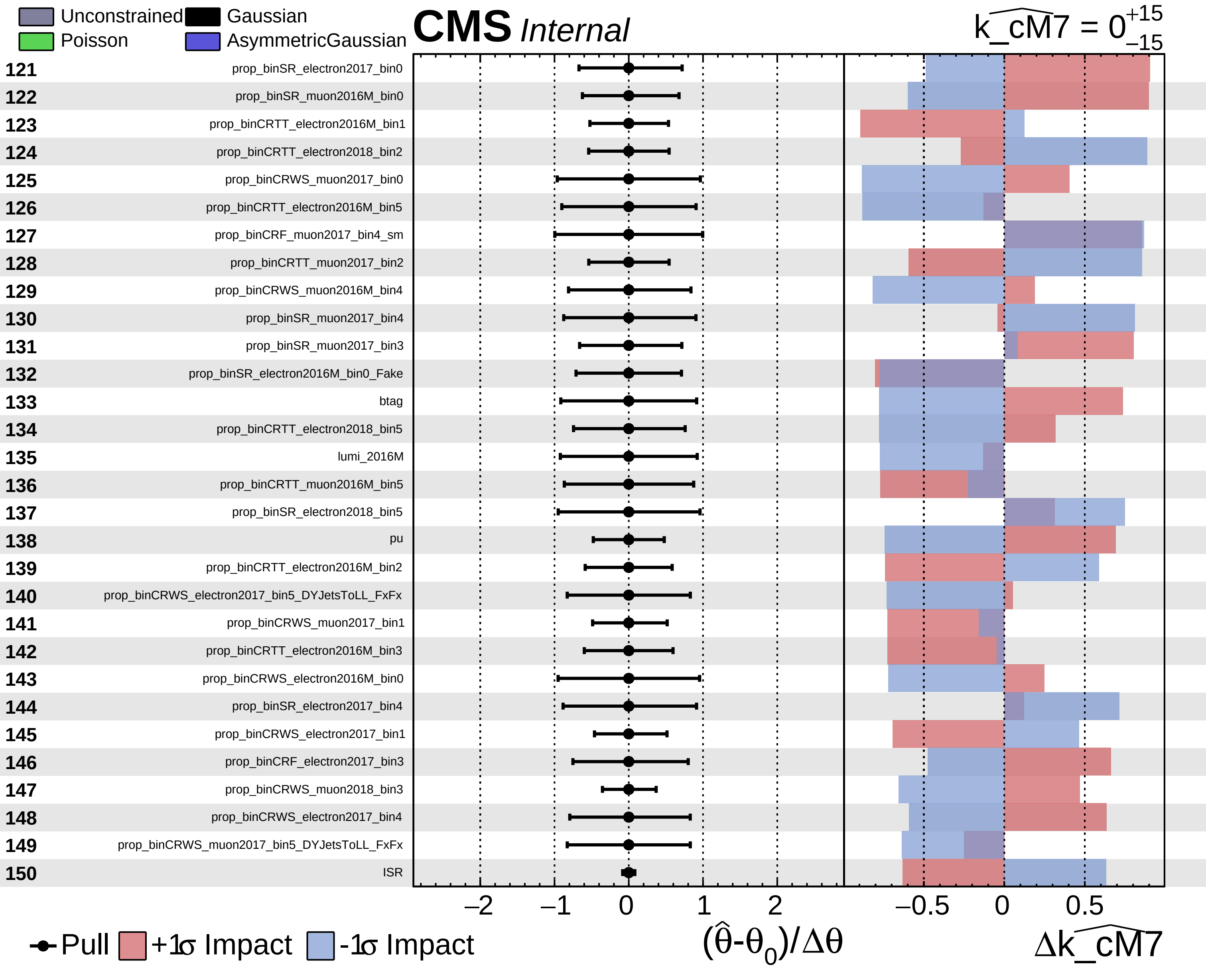
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +15 −15
31 prop_binCRWS_electron2018_bin1
32 prop_binSR_electron2016M_bin1
33 prop_binSR_muon2017_bin0
34 prop_binCRF_muon2017_bin1
35 prop_binSR_muon2018_bin1
36 prop_binCRF_electron2017_bin1
37 DYest_muon_2017
1.00 +0.24 −0.24
38 prop_binCRF_electron2017_bin0
39 prop_binCRWS_muon2017_bin2
40 prop_binSR_electron2018_bin2
41 FRest_muon_2016M
1.00 +0.12 −0.12
42 prop_binCRWS_electron2018_bin4
43 prop_binCRWS_electron2018_bin3
44 FRest_muon_2017
1.00 +0.12 −0.12
45 prop_binCRWS_electron2017_bin3
46 prop_binCRWS_electron2017_bin0
47 prop_binSR_muon2018_bin2
48 TTest_electron_2017
1.00 +0.07 −0.07
49 prop_binSR_muon2018_bin0
50 prop_binCRWS_muon2018_bin2
51 DYest_muon_2016M
1.00 +0.30 −0.30
52 prop_binCRWS_muon2018_bin4
53 prop_binCRWS_electron2018_bin5
54 prop_binCRTT_muon2016M_bin2
55 prop_binCRF_muon2016M_bin0
56 mistag
57 prop_binCRTT_electron2018_bin1
58 prop_binCRF_muon2017_bin2
59 prop_binCRTT_electron2018_bin3
60 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−2 −1 0 1 2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

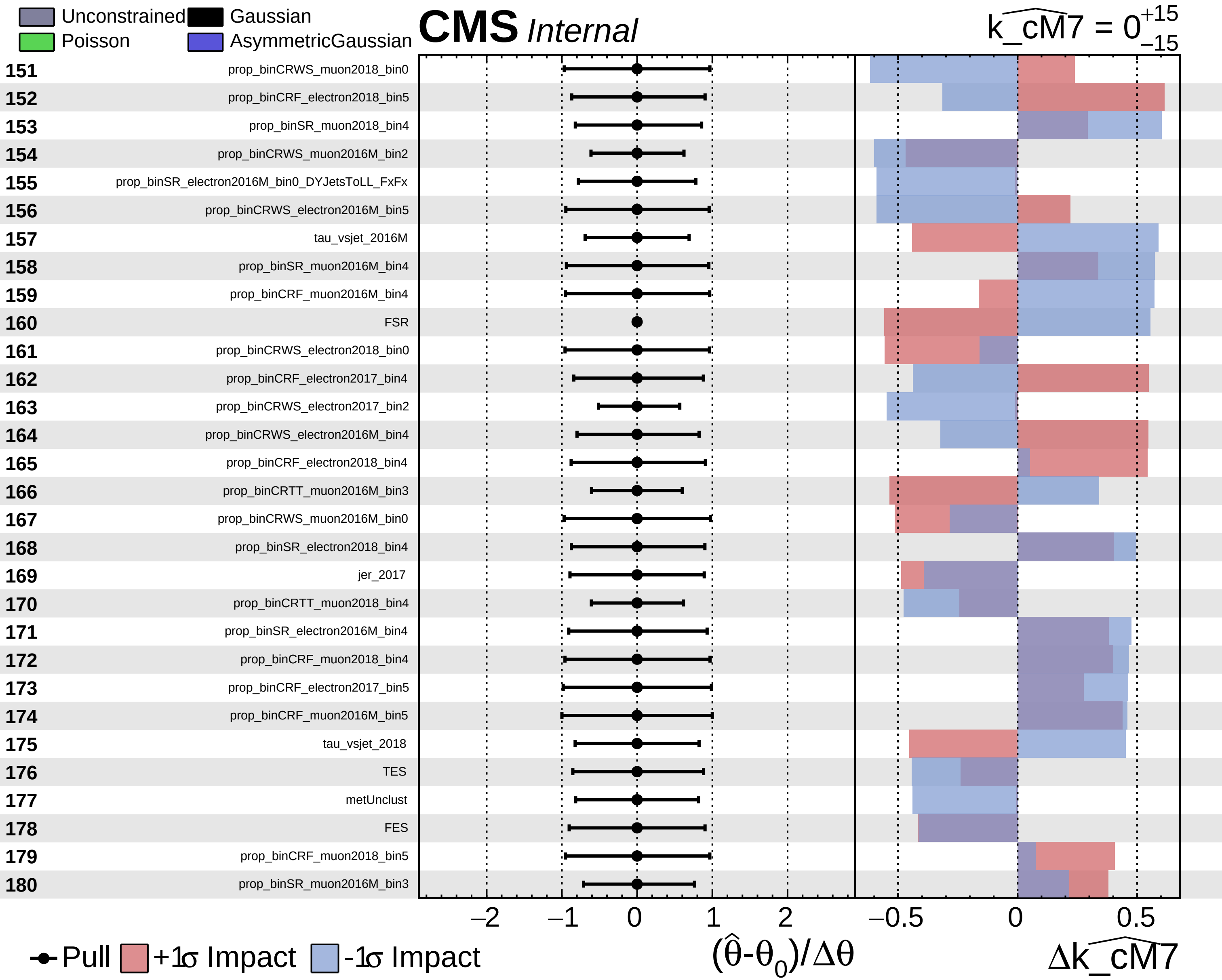


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +15 −15
61 prop_binCRF_electron2016M_bin0
62 prop_binCRTT_muon2017_bin5
63 prop_binCRWS_muon2017_bin3
64 prop_binCRF_electron2016M_bin3
65 TTest_electron_2018 1.00 +0.07 −0.07
66 FRest_muon_2018 1.00 +0.11 −0.11
67 prop_binCRF_electron2017_bin2
68 prop_binSR_muon2017_bin2
69 prop_binCRTT_muon2017_bin3
70 prop_binCRWS_electron2016M_bin1
71 FR_sys_electron_2018
72 TTest_electron_2016M 1.00 +0.09 −0.09
73 prop_binSR_electron2017_bin2
74 FRest_electron_2017 1.00 +0.12 −0.12
75 prop_binCRTT_muon2017_bin0
76 FRest_electron_2018 1.00 +0.11 −0.11
77 prop_binCRTT_muon2018_bin2
78 prop_binCRF_muon2016M_bin1
79 prop_binSR_muon2016M_bin1
80 prop_binSR_electron2018_bin0
81 prop_binCRF_electron2016M_bin2
82 lep_2017
83 jes
84 prop_binSR_muon2016M_bin2
85 prop_binSR_electron2018_bin3
86 prop_binCRTT_electron2017_bin5
87 lep_2018
88 prop_binSR_electron2017_bin1
89 prop_binCRF_muon2016M_bin2
90 prop_binCRTT_electron2017_bin4
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

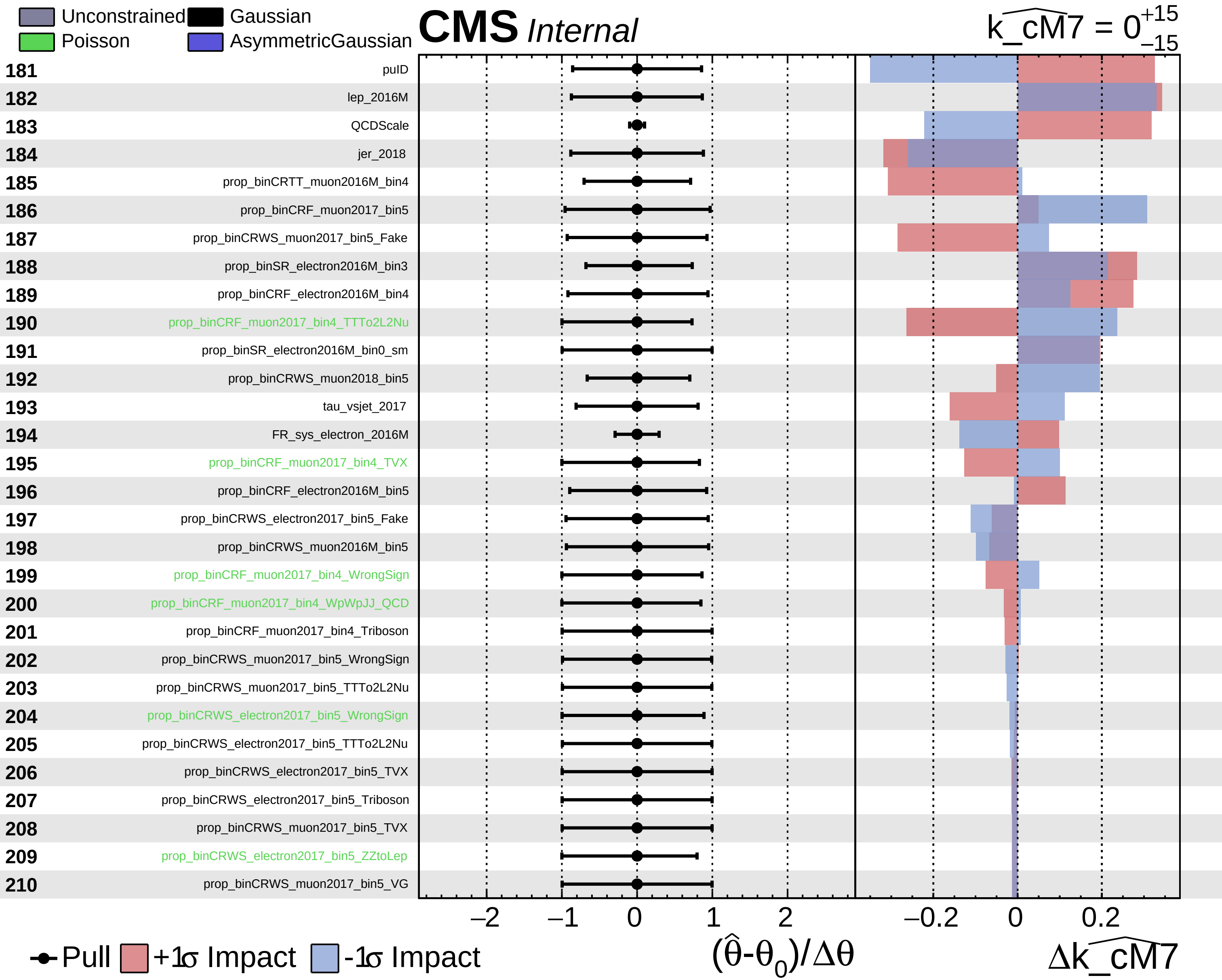


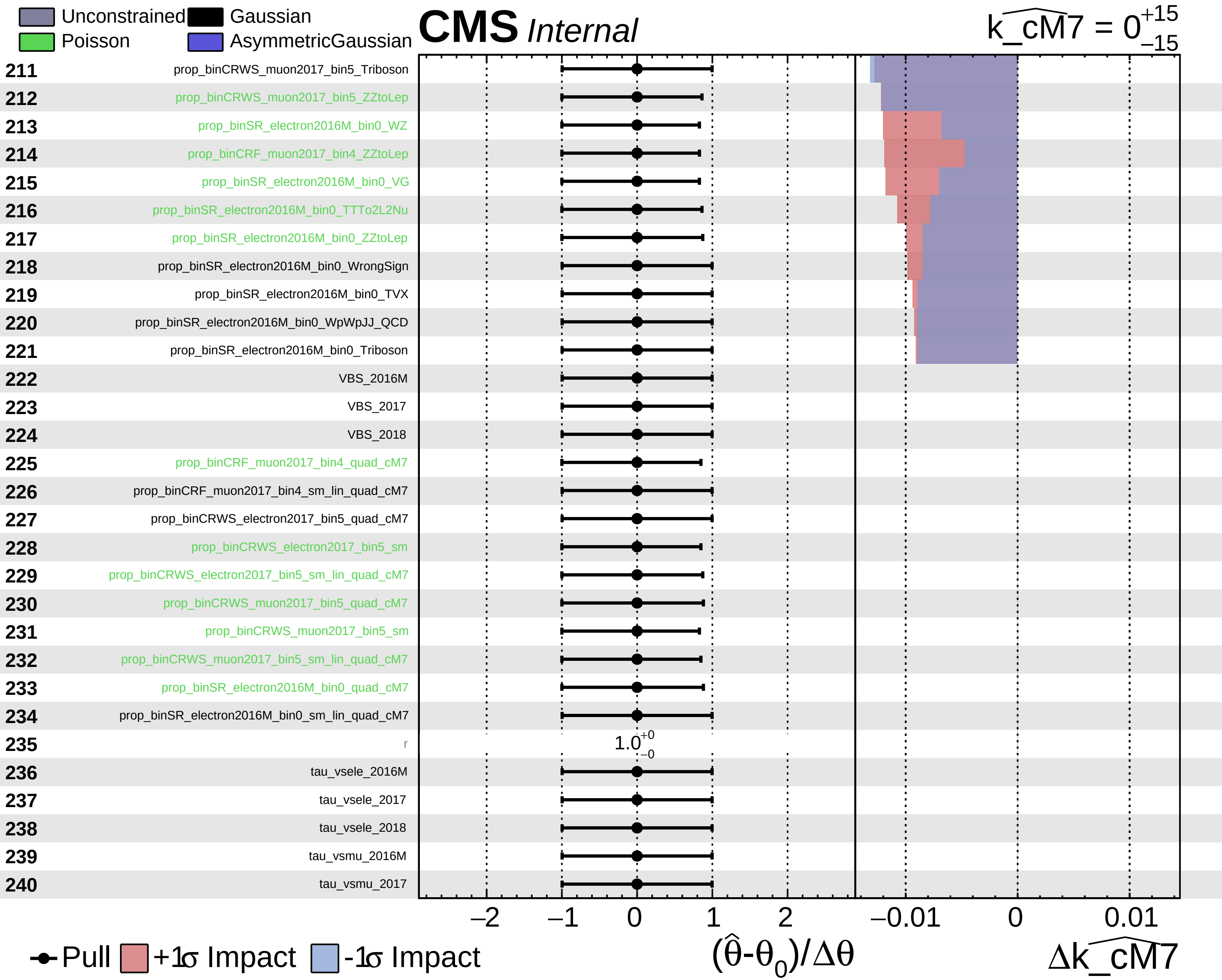
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+15}_{-15}$
91 prop_binSR_electron2017_bin3
92 FR_sys_electron_2017
93 prop_binCRTT_electron2016M_bin4
94 prop_binCRTT_muon2018_bin5
95 FR_sys_muon_2016M
96 PF
97 prop_binCRTT_muon2016M_bin1
98 prop_binCRTT_muon2017_bin1
99 prop_binCRTT_muon2018_bin3
100 prop_binCRF_muon2016M_bin3
101 prop_binCRF_muon2018_bin3
102 prop_binCRWS_muon2017_bin4
103 prop_binCRWS_muon2018_bin1
104 FR_sys_muon_2018
105 prop_binSR_electron2016M_bin2
106 DYest_electron_2017
$1.00^{+0.31}_{-0.31}$
107 prop_binCRF_muon2017_bin3
108 prop_binCRTT_electron2016M_bin0
109 prop_binCRTT_electron2018_bin4
110 FRest_electron_2016M
$1.00^{+0.12}_{-0.12}$
111 lumi_2017
112 prop_binCRTT_muon2017_bin4
113 jer_2016M
114 prop_binCRWS_muon2016M_bin1
115 prop_binSR_electron2016M_bin5
116 prop_binCRWS_electron2016M_bin2
117 prop_binCRWS_electron2018_bin2
118 prop_binCRWS_electron2016M_bin3
119 lumi_2018
120 FR_sys_muon_2017
−2 −1 0 1 2
−1 0 1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +15 −15
121 prop_binSR_electron2017_bin0
122 prop_binSR_muon2016M_bin0
123 prop_binCRTT_electron2016M_bin1
124 prop_binCRTT_electron2018_bin2
125 prop_binCRWS_muon2017_bin0
126 prop_binCRTT_electron2016M_bin5
127 prop_binCRF_muon2017_bin4_sm
128 prop_binCRTT_muon2017_bin2
129 prop_binCRWS_muon2016M_bin4
130 prop_binSR_muon2017_bin4
131 prop_binSR_muon2017_bin3
132 prop_binSR_electron2016M_bin0_Fake
133 btag
134 prop_binCRTT_electron2018_bin5
135 lumi_2016M
136 prop_binCRTT_muon2016M_bin5
137 prop_binSR_electron2018_bin5
138 pu
139 prop_binCRTT_electron2016M_bin2
140 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
141 prop_binCRWS_muon2017_bin1
142 prop_binCRTT_electron2016M_bin3
143 prop_binCRWS_electron2016M_bin0
144 prop_binSR_electron2017_bin4
145 prop_binCRWS_electron2017_bin1
146 prop_binCRF_electron2017_bin3
147 prop_binCRWS_muon2018_bin3
148 prop_binCRWS_electron2017_bin4
149 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
150 ISR
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7

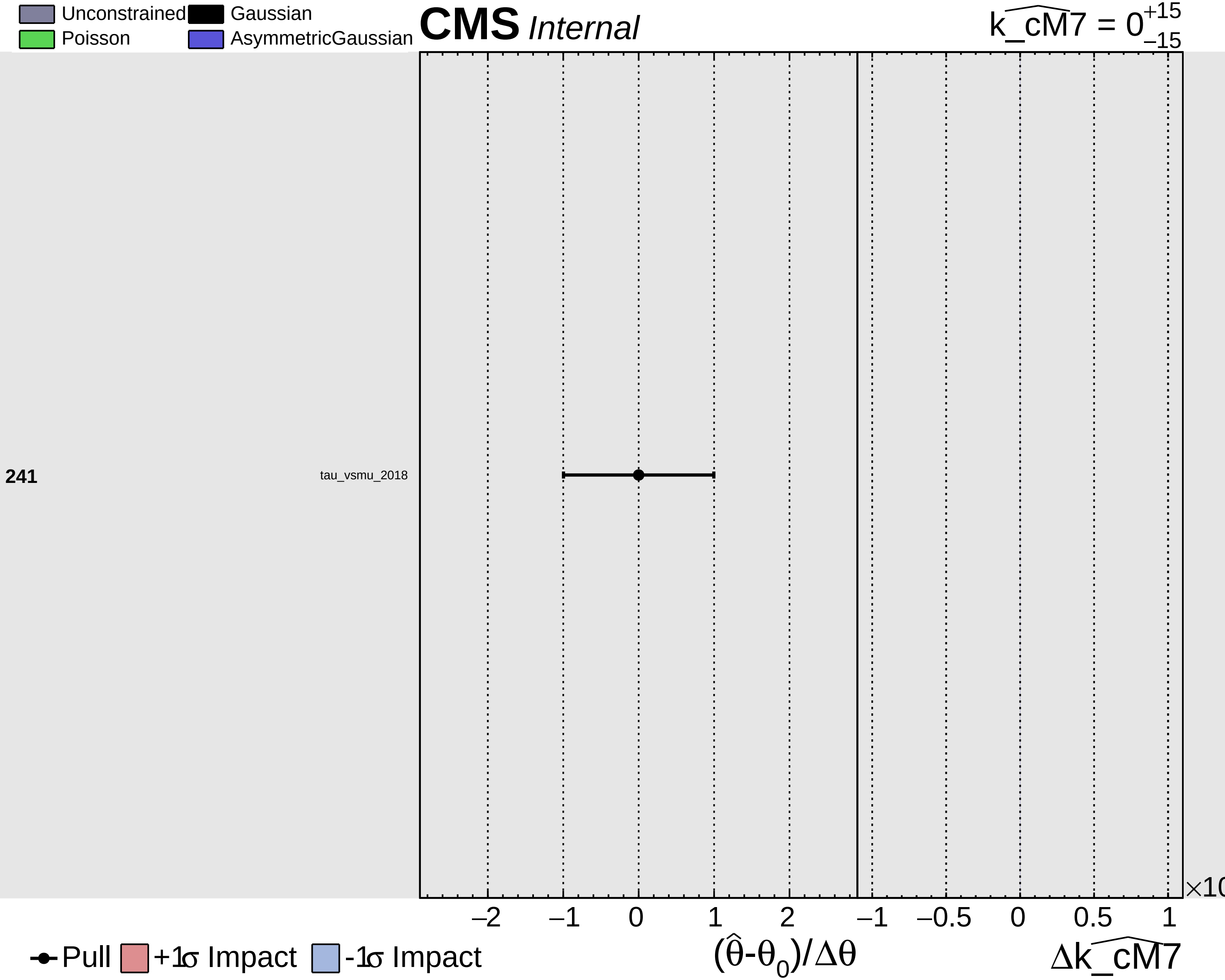


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +15 −15
151 prop_binCRWS_muon2018_bin0
152 prop_binCRF_electron2018_bin5
153 prop_binSR_muon2018_bin4
154 prop_binCRWS_muon2016M_bin2
155 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
156 prop_binCRWS_electron2016M_bin5
157 tau_vsjet_2016M
158 prop_binSR_muon2016M_bin4
159 prop_binCRF_muon2016M_bin4
160 FSR
161 prop_binCRWS_electron2018_bin0
162 prop_binCRF_electron2017_bin4
163 prop_binCRWS_electron2017_bin2
164 prop_binCRWS_electron2016M_bin4
165 prop_binCRF_electron2018_bin4
166 prop_binCRTT_muon2016M_bin3
167 prop_binCRWS_muon2016M_bin0
168 prop_binSR_electron2018_bin4
169 jer_2017
170 prop_binCRTT_muon2018_bin4
171 prop_binSR_electron2016M_bin4
172 prop_binCRF_muon2018_bin4
173 prop_binCRF_electron2017_bin5
174 prop_binCRF_muon2016M_bin5
175 tau_vsjet_2018
176 TES
177 metUnclust
178 FES
179 prop_binCRF_muon2018_bin5
180 prop_binSR_muon2016M_bin3
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+15}_{-15}$
181 puID
182 lep_2016M
183 QCDScale
184 jer_2018
185 prop_binCRTT_muon2016M_bin4
186 prop_binCRF_muon2017_bin5
187 prop_binCRWS_muon2017_bin5_Fake
188 prop_binSR_electron2016M_bin3
189 prop_binCRF_electron2016M_bin4
190 prop_binCRF_muon2017_bin4_TTTo2L2Nu
191 prop_binSR_electron2016M_bin0_sm
192 prop_binCRWS_muon2018_bin5
193 tau_vsjet_2017
194 FR_sys_electron_2016M
195 prop_binCRF_muon2017_bin4_TVX
196 prop_binCRF_electron2016M_bin5
197 prop_binCRWS_electron2017_bin5_Fake
198 prop_binCRWS_muon2016M_bin5
199 prop_binCRF_muon2017_bin4_WrongSign
200 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
201 prop_binCRF_muon2017_bin4_Triboson
202 prop_binCRWS_muon2017_bin5_WrongSign
203 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
204 prop_binCRWS_electron2017_bin5_WrongSign
205 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
206 prop_binCRWS_electron2017_bin5_TVX
207 prop_binCRWS_electron2017_bin5_Triboson
208 prop_binCRWS_muon2017_bin5_TVX
209 prop_binCRWS_electron2017_bin5_ZZtoLep
210 prop_binCRWS_muon2017_bin5_VG
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0^{+15}_{−15}
211 prop_binCRWS_muon2017_bin5_Triboson
212 prop_binCRWS_muon2017_bin5_ZZtoLep
213 prop_binSR_electron2016M_bin0_WZ
214 prop_binCRF_muon2017_bin4_ZZtoLep
215 prop_binSR_electron2016M_bin0_VG
216 prop_binSR_electron2016M_bin0_TTTo2L2Nu
217 prop_binSR_electron2016M_bin0_ZZtoLep
218 prop_binSR_electron2016M_bin0_WrongSign
219 prop_binSR_electron2016M_bin0_TVX
220 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
221 prop_binSR_electron2016M_bin0_Triboson
222 VBS_2016M
223 VBS_2017
224 VBS_2018
225 prop_binCRF_muon2017_bin4_quad_cM7
226 prop_binCRF_muon2017_bin4_sm_lin_quad_cM7
227 prop_binCRWS_electron2017_bin5_quad_cM7
228 prop_binCRWS_electron2017_bin5_sm
229 prop_binCRWS_electron2017_bin5_sm_lin_quad_cM7
230 prop_binCRWS_muon2017_bin5_quad_cM7
231 prop_binCRWS_muon2017_bin5_sm
232 prop_binCRWS_muon2017_bin5_sm_lin_quad_cM7
233 prop_binSR_electron2016M_bin0_quad_cM7
234 prop_binSR_electron2016M_bin0_sm_lin_quad_cM7
235 r
1.0^{+0}_{−0}
236 tau_vsele_2016M
237 tau_vsele_2017
238 tau_vsele_2018
239 tau_vsmu_2016M
240 tau_vsmu_2017
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+15}_{-15}$
241
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$